Phase 2 and Poster — preparation guide

IASTAM 6.0 • Track 4 • Problem 7 — 24 September 2026. Companion to the technical report: what to deliver for Phase 2, how to build the poster, and what to do about SAHI.

1. Where we stand today

Piece	State	Evidence
Problem understanding + literature	Done	12 papers read, limitations verified in the text
Architecture and design	Done	Pipeline, level of detail, scheduler formulation
Code base + tests	Done	28 automated tests pass
Contact windows [REAL]	Done	5 passes/day over Sfax, 11.4 h gap
Scheduler comparison [SIM]	Done	68.5% → 93.0% from scheduling alone
Data integrity (WP0) [REAL]	Done	53,195 tiles, 6,282 duplicate pairs, 0 leakage
Detector (WP1)	Not started	Dataset is ready to train
Calibration (WP2), measured LoD (WP4)	Not started	Phase 3
Pitch video, interim paper	Not started	Due 26 September

The gap that matters: we have no detector numbers yet, and the jury's biggest block of points is “actual prototype progress”. Everything below is organised around closing that gap in three days.

2. What Phase 2 actually requires

Deliverable	Rule (spec book)	Our plan
Pitch video	Maximum 2 minutes: problem, core idea, technical approach	1 take, screen recording + voice-over, no fancy editing
Interim research paper	Academic conventions, IEEE double column recommended; title/authors, abstract 150–250 words + 3–5 keywords, I Introduction, II Related Work, III Methodology, IV Experiments & Roadmap, V Conclusion, References	Sections II and III are already written in the technical report
Mid-review	Scored out of 100; ≥ 60 required to continue	See the scoring map below
Poster session	Open to all teams, even those not selected for the final pitch; templates and guidelines shared by the organisers after registration	Build content now, drop it into the official template when it arrives

3. Three-day plan (23 → 26 September)

When	Task	Owner	Output
Wed 23, evening	Launch YOLOv8n training on Kaggle/Colab (dataset is ready)	Detector	Training running overnight
Wed 23	Paper skeleton in Overleaf, paste Related Work + Methodology from the report	Paper	Sections I–III drafted
Thu 24	Detector results: mAP50, recall by ship size, FP32 vs INT8, CPU latency	Detector	predictions.csv + table
Thu 24	Feed real detections into the scheduler; rerun the comparison	You	Real-data version of the main chart
Thu 24	Fair fixed-patch baseline (same coastal handling)	Scheduler	Honest comparison
Fri 25	Finish paper, insert figures, proofread; record the video	Paper + all	PDF + MP4
Fri 25	Poster content block by block (see section 5)	Visuals	Poster draft
Sat 26	Buffer, final checks, submit before the deadline	All	Submitted

Rule for these three days: no new features. Only measurements, writing and rehearsal. A working small result beats an unfinished big one.

4. The 2-minute video and the paper

Time	What is on screen	What you say
0:00–0:20	One satellite image, then the same image reduced to a few numbers	The problem: a satellite sees far more than it can send; 5 passes a day, 34 minutes of contact
0:20–0:50	The 6-step diagram, steps 4–5 highlighted	Our idea: send information about ships, and decide how many bytes each ship deserves
0:50–1:20	Architecture + level-of-detail table	How it works: cheap filter, detector, confidence decides the detail level
1:20–1:45	The main result chart	What we measured, and honestly what is simulated
1:45–2:00	Roadmap slide	What comes before the final: real detector, calibration, demo

- Record the screen with your voice; no music, no slow intro animation — the first 10 seconds decide attention.
- Say a number in the first 20 seconds (11.4 h between contacts).
- State clearly which results are simulated: juries trust teams that separate measured from expected.
- Keep a 10 s safety margin: a 2:05 video can be rejected.

Paper: where each part already exists

Section	Source in our material	Extra work
Abstract + keywords	Executive summary of the technical report	Rewrite to 150–250 words
I. Introduction	Report sections 1–2 (bottleneck, real passes)	Add our contributions as a list
II. Related Work	Report section 2 table with stated limitations	Turn the table into prose
III. Methodology	Report section 3 + WP0 method	Add the architecture figure
IV. Experiments & Roadmap	Report section 5 + detector results (Thursday)	Add the WP plan table
V. Conclusion	Report section 7	Two paragraphs
References	Report appendix C	Convert to IEEE style

Best Scientific Paper award: what wins it is method honesty, not volume — the leakage-free split, the calibration story, confidence intervals, and the table of our own weaknesses. We already have all four.

5. The poster

The organisers will send an official template; build the content now and drop it in. Standard format is A0 portrait (841 × 1189 mm). A poster is read from 1.5–2 m away, in a noisy room, in about 60 seconds.



Suggested layout: three columns, one message per block. Numbers in the blocks refer to the reading order.

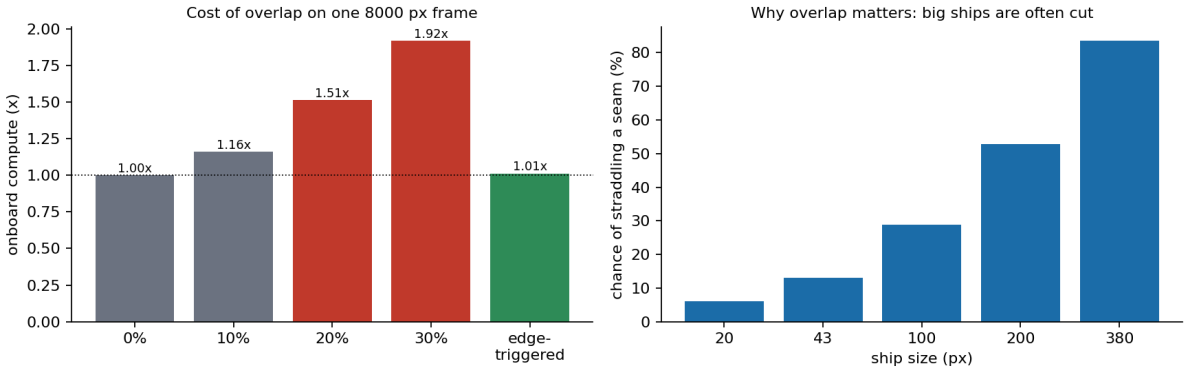
Rules that decide whether it is read

Do	Don't
One sentence as the main message, in the header	A wall of paragraphs
Figures ≥ 40% of the surface	Screenshots of code or terminal output
Body text ≥ 24 pt, headings ≥ 36 pt, title ≥ 72 pt	Text below 20 pt — unreadable standing up
Maximum ~800 words in total	Full paper sections copied in
Big numbers as headlines (93%, 11.4 h, 19%)	Tables with 20 rows
Mark clearly what is simulated	Presenting targets as results
QR code to the repository and the report	URLs written out by hand
Print 48 h early; bring a PDF backup on a USB key	Printing the morning of the congress

Prepare a **60-second spoken version** and a **5-minute version**. Most visitors get the first; the jury gets the second.

6. SAHI: should we use it?

SAHI (Slicing Aided Hyper Inference) cuts a large frame into overlapping tiles so an object on a tile border is not split in half, then merges the boxes. The overlap is what costs us.



Left: onboard compute for one 8000 px frame with 640 px tiles. Right: probability that a ship straddles a tile border, by ship size (Airbus boxes reach 380 px).

Finding	Number	Consequence
Standard 20% overlap	169 → 256 tiles = 1.51× compute	Expensive where energy is the constraint
Median ship (43 px)	13% chance of being cut	Moderate problem
Large ships (200–380 px)	53–84% chance of being cut	Serious: the most valuable targets
Our data today	Airbus tiles are already 768 px	SAHI changes nothing for the current results

Our alternative: edge-triggered re-inference

Instead of paying 51% more compute everywhere, spend it only where a ship might be cut. The cheap classic pre-filter already finds bright blobs; if a blob or a detection *touches a tile border*, run one extra inference on a window centred on that seam. Cost measured on a typical frame: about **1.01×** instead of 1.51×.

- It reuses WP3 (the classic filter) to decide where WP1 (the detector) must look again — the two parts of our system cooperate.
- It is measurable: recall of ships near seams, and compute cost, for three policies (no overlap, SAHI 20%, edge-triggered).
- It is an honest comparison against a well-known baseline (SAHI raised AP by 5–7 points on aerial datasets), which is exactly what a jury wants to see.

Timing: this is Phase 3 work, not before the mid-review. Mention it in the paper's roadmap as planned work, with the cost numbers above — they already show we understand the trade-off.

Experiment protocol (Phase 3)

Step	How
Build the test	Take Airbus tiles and slice each into 2×2 sub-tiles: ships on the seams are known from the labels
Policies	(a) no overlap, (b) SAHI 20% overlap + box merging, (c) edge-triggered re-inference
Measure	Recall of seam ships, recall overall, number of inferences, runtime
Report	One chart: recall versus compute cost for the three policies

7. Scoring map: where the 100 points come from

Criterion	Pts	What we show	Status
Understanding of the problem	15	Real contact windows, verified literature limitations	Strong
Relevance and originality	15	Confidence-aware level of detail + multi-pass scheduler	Strong
Technical architecture	20	Pipeline, interfaces, design patterns	Strong
Feasibility and realism	15	Realistic link, COTS evidence, FMEA	Strong
Actual prototype progress	20	Code, 28 tests, WP0 on real data, detector (this week)	Needs WP1
Methodology and planning	10	Leakage-free split, baselines, confidence intervals	Strong
Presentation	5	Video, poster, rehearsed answers	To do

8. Questions to rehearse

- Which of your results are measured and which are simulated?
- What happens when the detector misses a ship?
- Why greedy scheduling, and how far from optimal is it?
- How do you know your test scores are not inflated by duplicate images?
- Why not simply compress the images better?
- What does “no AIS signal” actually prove?
- Is your comparison against Φ-sat-2 fair?
- What would you do differently with a real satellite?

Every one of these has an answer in the technical report; make sure each team member can give it in 30 seconds.